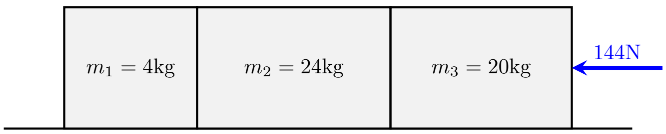
Dynamics Test Answers

Ethan Wang

April 28, 2026

Blocks(60 points)

1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $3\,\mathrm{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $3\,\mathrm{ms}^{-2}$ left

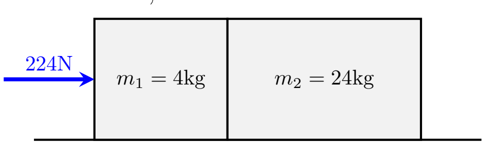
(c) The net force on each block

Solution: Block 1 has a net force of 12N left
Block 2 has a net force of 72N left
Block 3 has a net force of 60N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
12N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
12N right (contact force from block 1), 84N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
144N left (applied force), 84N right (contact force from block 2)

2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $8\,\mathrm{ms}^{-2}$ right

(b) The acceleration of each block

Solution: Each block has an acceleration of $8\,\mathrm{ms^{-2}}$ right

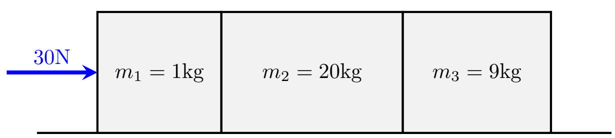
(c) The net force on each block

Solution: Block 1 has a net force of 32N right
Block 2 has a net force of 192N right

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
224N right (applied force), 192N left (contact force from block 2)
Block 2 has 1 force acting on it. They are:
192N right (contact force from block 1)

3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $1\,\mathrm{ms^{-2}}$ right

(b) The acceleration of each block

Solution: Each block has an acceleration of $1\,\mathrm{ms^{-2}}$ right

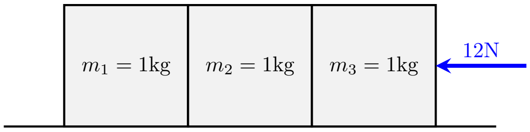
(c) The net force on each block

Solution: Block 1 has a net force of 1N right
Block 2 has a net force of 20N right
Block 3 has a net force of 9N right

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
30N right (applied force), 29N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
29N right (contact force from block 1), 9N left (contact force from block 3)
Block 3 has 1 force acting on it. They are:
9N right (contact force from block 2)

4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $4\,\text{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $4\,\text{ms}^{-2}$ left

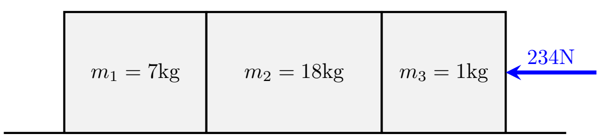
(c) The net force on each block

Solution: Block 1 has a net force of 4N left
Block 2 has a net force of 4N left
Block 3 has a net force of 4N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
4N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
4N right (contact force from block 1), 8N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
12N left (applied force), 8N right (contact force from block 2)

5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $9\,\text{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $9\,\text{ms}^{-2}$ left

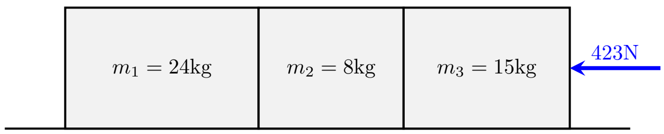
(c) The net force on each block

Solution: Block 1 has a net force of 63N left
Block 2 has a net force of 162N left
Block 3 has a net force of 9N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
63N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
63N right (contact force from block 1), 225N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
234N left (applied force), 225N right (contact force from block 2)

6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $9\,\mathrm{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $9\,\mathrm{ms}^{-2}$ left

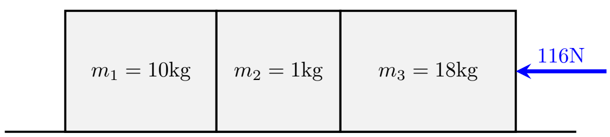
(c) The net force on each block

Solution: Block 1 has a net force of 216N left
Block 2 has a net force of 72N left
Block 3 has a net force of 135N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
216N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
216N right (contact force from block 1), 288N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
423N left (applied force), 288N right (contact force from block 2)

7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $4\,\mathrm{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $4\,\mathrm{ms}^{-2}$ left

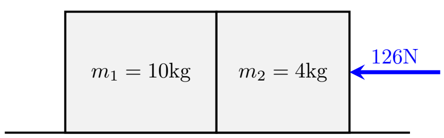
(c) The net force on each block

Solution: Block 1 has a net force of 40N left
Block 2 has a net force of 4N left
Block 3 has a net force of 72N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
40N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
40N right (contact force from block 1), 44N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
116N left (applied force), 44N right (contact force from block 2)

8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $9\,\mathrm{ms}^{-2}$ left

(b) The acceleration of each block

Solution: Each block has an acceleration of $9\,\mathrm{ms}^{-2}$ left

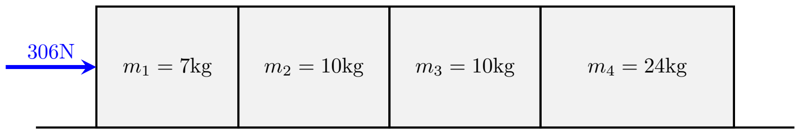
(c) The net force on each block

Solution: Block 1 has a net force of 90N left
Block 2 has a net force of 36N left

(d) The contact forces for each block

Solution: Block 1 has 1 force acting on it. They are:
90N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
126N left (applied force), 90N right (contact force from block 1)

9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $6\,\mathrm{ms}^{-2}$ right

(b) The acceleration of each block

Solution: Each block has an acceleration of $6\,\mathrm{ms}^{-2}$ right

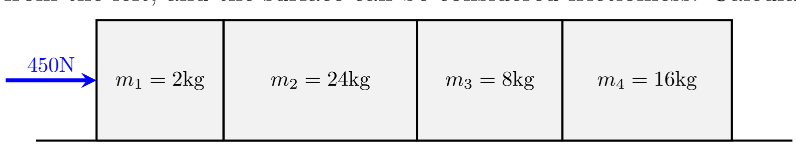
(c) The net force on each block

Solution: Block 1 has a net force of 42N right
Block 2 has a net force of 60N right
Block 3 has a net force of 60N right
Block 4 has a net force of 144N right

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
306N right (applied force), 264N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
264N right (contact force from block 1), 204N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
204N right (contact force from block 2), 144N left (contact force from block 4)
Block 4 has 1 force acting on it. They are:
144N right (contact force from block 3)

10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



(a) The acceleration of the system

Solution: The acceleration of the system is $9\,\mathrm{ms}^{-2}$ right

(b) The acceleration of each block

Solution: Each block has an acceleration of $9\,\mathrm{ms}^{-2}$ right

(c) The net force on each block

Solution: Block 1 has a net force of 18N right
Block 2 has a net force of 216N right
Block 3 has a net force of 72N right
Block 4 has a net force of 144N right

(d) The contact forces for each block

Solution: Block 1 has 2 forces acting on it. They are:
450N right (applied force), 432N left (contact force from block 2)
Block 2 has 2 forces acting on it. They are:
432N right (contact force from block 1), 216N left (contact force from block 3)
Block 3 has 2 forces acting on it. They are:
216N right (contact force from block 2), 144N left (contact force from block 4)
Block 4 has 1 force acting on it. They are:
144N right (contact force from block 3)